Group Name : False 9

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Question 1

a) $S = \{E, S, P, D\}$

b) i) $A = \{E, D\}$

$B = \{S\}$

$A \cup B = \{E, D, S\}$

ii) $A \cap B = \{\}$

c) i) $P(E) = P(S) = P(P) = P(D) = 0.250$

$P(A) = 0.250 + 0.250$

$= 0.500$ ※

ii) $P(B) = 0.250$

$P(B') = 1 - 0.250$

$= 0.750$ ※

Question 2

a) $0.10 + 0.25 + 0.30 + 0.20 + a = 1$

$a = 1 - 0.85$

$a = 0.15$

b) $P(X=0) + P(X=1) + P(X=2)$

i) $= 0.10 + 0.25 + 0.30$

$= 0.65$

ii) $P(1 < X \leq 3) = P(X=2) + P(X=3)$

$= \cancel{0.25} + \cancel{0.30}\ 0.30 + 0.20$

$= 0.50$

c) mean $= 0(0.10) + 1(0.25) + 2(0.30)$

$+ 3(0.20) + 4(0.15)$

$= 2.05$

variance $= 0^2(0.10) + 1^2(0.25) + 2^2(0.30) + 3^2(0.20)$

$+ 4^2(0.15)$

$= 5.65$

$\delta^2 = 5.65 - (2.05)^2$

$= 1.448$

Question 3

a) $X = \{0,1,2,3,4\}$

b) $X \sim B(4, 0.20)$

c) i) $P(X=2) = {}^4C_2\,(0.2)^2(0.8)^2$

$= 0.154$

ii) $P(X \leq 1) = P(X=0) + P(X=1)$

$= {}^4C_0\,(0.2)^0(0.8)^4 + {}^4C_1\,(0.2)^1(0.8)^3$

$= 0.819$

d) mean $= (4)(0.2)$

$= 0.8$

variance $= (4)(0.2)(0.8)$

$= 0.64$

Question 4

a) $Y = \{1, 2, 3, 4, 5 \ldots\}$; all integer variable

b) $Y \sim Geo(p)$

c) $P(Y=5) = (1-0.10)^{5-1}(0.10)$

i) $= 0.066$

ii) $P(Y \leq 4) = 1 - P(Y > 4)$

$= 1 - (1-0.10)^4$

$= 0.344$

Question 5

$\mu = 40$ minutes ; $z = \dfrac{x-\mu}{\sigma}$

$\sigma = 6$ minutes

a)(i) $P(X<46)$

$$z = \frac{46-40}{6} = 1.00$$

$P(z<1.00) = 0.8413$

$P(X<46) = \underline{0.8413}$

b) $P(X>x) = 0.05$

$P(X<x) = 0.95$

z-table $(0.95) = 1.645$

$x = z\sigma + \mu$

$= \cancel{4}(1.645)(6) + 40$

$= \underline{49.87}$

(ii) $P(34<X<52)$

$x = 34$;

$$z = \frac{34-40}{6}$$

$= -1.00$

$P(z<-1.00) = 0.1587$

$x = 52$

$$z = \frac{52-40}{6} = 2.00$$

$P(z<2.00) = 0.9772$

$P(34<X<52) = 0.9772 - 0.1587$

$= \underline{0.8185}$

Question 6

$p = 0.6$; $q = 0.4$; $r = 3$

(a) $X \sim NB(r=3, p=0.60)$

(b) $\sigma = \sqrt{\dfrac{rq}{p^2}} = \sqrt{\dfrac{3(0.4)}{(0.6)^2}}$

$= \underline{1.826}$

(c) negative binomial

$$P(X=x) = \binom{x-1}{r-1} p^r q^{x-r}$$

$$P(X=5) = \binom{5-1}{3-1} (0.60)^3 (0.40)^{5-3}$$

$= \underline{0.20736}$